

Technical Samples

In accordance with the solicitation and Statement of Work (SOW) please fabricate a removable partial denture framework, a fully balanced complete denture set-up and a porcelain fused to metal (PFM) fixed partial denture. You will be requested to pick-up a partial denture cast, mounted edentulous cast with record bases and occlusion rims and a cast for a 3 unit PFM fixed partial denture. **Please follow the enclosed laboratory prescriptions** to fabricate the removable partial denture framework, complete denture set up and the PFM fixed partial denture.

You will have 15 days to complete lab work and deliver to the Birmingham VA Medical Center, Room 2625, after notification from the contracting officer. This will take place AFTER the completion of Phase I, as outlined in the solicitation. At the time of pick-up and delivery an e-mail notification shall be sent to the Contracting Officer at aunjelaca.miller@va.gov as soon as possible, notifying the action.

*Attention: Dental Service Administrative Officer, Rm 2625
Birmingham VA Medical Center
700 South 19th Street
Birmingham, Al. 35223*

*Primary POC: Sonni Woodbury
Sonni.Woodbury@va.gov
(205) 933-8101, ext. 336466
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*Alternate POC: Jimmy Reed
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205-933-8101*

See below for more specific guidelines for how the prostheses will be evaluated for quality.

(From Statement of Work)

Qualifying dental laboratories will submit the following lab work samples to be evaluated by the Birmingham VA Dental Laboratory (BVDL) as requested by the solicitation:

1. **Removable partial denture framework:** A cast will be provided by the BVDL. Lab work will be fabricated according to laboratory prescription and evaluated as stated in the statement of work.
2. **Complete denture set-up**—BVDL will provide mounted cast with record bases and wax rims. Laboratory will provide a cross-tooth cross arch fully balanced denture set-up

completely waxed and ready for investing. Lab work will be fabricated according to laboratory prescription and evaluated as stated in the statement of work.

3. **Anterior fixed partial denture**—BVDL will provide a cast of a fixed partial denture. Laboratory will pin, saw, trim dies and mount on provided semi-adjustable articulator using bite registration and provided opposing cast. Lab will fabricate porcelain fused to high noble metal fixed partial denture. Lab work will be fabricated according to laboratory prescription and evaluated as stated in the statement of work.

QUALITY ASSESSMENT GUIDELINES

I. Complete and Partial Dentures

1. Denture and partial denture impression must be boxed and poured in improved stone and be dense and free from voids. Denture and partial cast must have a peripheral roll on the cast such that the border molding has been maintained. The roll should be 2-3 mm in height, and the land area should also be 2-3 mm wide for the border molded area.

2. Denture acrylic must be free from porosity and highly polished. Ethnic acrylic shades must be available for use and ethnic shade guides must be provided to the VA lab to help in selecting the proper acrylic shade for the patient. Borders must be maintained when processing and finishing prostheses. There should be no attempt to fit processed denture or partial denture bases back to the master cast. Acrylic around denture teeth shall be anatomically correct and maintained after polishing with a minimal amount of flash acrylic on denture teeth.

3. Lab must be able to set different types of occlusal schemes as requested to include:

- a. Cross tooth cross arch balanced occlusal scheme
- b. Lingualized occlusal scheme
- c. Lingualized non-balanced occlusal scheme
- d. Monoplane/neutrocentric occlusal scheme

REF-1. Dental Laboratory Procedures---Morrow, Rudd and Eissmann

2. U.S. Air Force Lab Manual—AFPAM-103VI

4. All denture and partial denture cases shall be remounted before separating from the master cast to insure pin opening from processing is less than 1mm and also to correct any occlusal discrepancy. For cases with pin opening greater than 1 mm

they must be adjusted at the lab. VA lab has the right to reject denture and partial denture cases whose teeth have been excessively adjusted and no longer resemble teeth.

5. All partial denture cast frameworks shall be returned on the master cast. The master cast shall not be abraded in any way. All rests must be completely seated within 75-100 microns and fit the full extent of the prepared rest seat. All frameworks must be smoothly polished and free from porosity. A nickel free alloy must be used in fabrication of metal partial denture frameworks.

6. All laboratory prescriptions done by VA dentists must be exactly followed. If a deviation from the requested design is required, the contract lab must request authorization from VA dentist.

7. Lab must provide an assortment of denture teeth that are available for use. Only the highest quality denture teeth shall be used in all prostheses.

8. All removable prostheses must have the patients' last name placed in the acrylic.

9. Removable cases mounted at the contract lab will be mounted on a calibrated whip mix 2240 articulator.

10. Excessive block-out shall not be used on record bases and every attempt shall be made to maintain the same peripheral border on the master cast in the record bases.

II. Crown and Bridge and Implant Prostheses

1. Microscopes must be used to evaluate all crowns and bridges and must fit the dies or implant custom abutments (within 75 microns). There shall be no ledges. The VA shall return any prostheses that do not meet these standards for repair or remake at contract lab expense.

2. All metal used for metal ceramic and gold crowns will be high noble. (ADA specifications for high noble-60% high noble metal of which 40% must be gold). The only time a base metal may be used is for resin retained fixed partial dentures (Maryland Bridges) or cast post and cores as requested. There will be no holes in metal copings on which porcelain has been applied.

3. All crowns and bridges should be anatomically correct with proper emergence profile and not over-contoured unless specified on the laboratory prescription.

4. All porcelain must be dense with no porosity and match precisely the requested shade tab, photograph and/or lab prescription custom design. No metal or opaque can be showing through areas where porcelain is supposed to be applied. All frameworks supporting the porcelain shall support the veneering porcelain such that there must be no more than 2 mm of unsupported veneering porcelain. VA lab must be able to add contact and occlusion if necessary to the applied porcelain. Contract lab will provide firing cycle numbers and sample material necessary to make chair-side corrections of the porcelain. If porcelain cracks/fractures within 1 year of insertion, contract laboratory will remake prosthesis at no cost to VA.
5. All cases should hold shim-stock when articulated. Occlusal contacts shall be on proper cusp, fossa and marginal ridges where possible. There shall be contact only in centric occlusion and not in lateral movements except where requested.
6. Cases mounted by contract lab will be mounted on a calibrated 2240 whip mix articulator.
7. No plastic articulators will be used for dual arch impressions. All cases must be mounted on a metal articulator.
8. Implant custom abutments must be fabricated according to the dentist specifications. Soft tissue cast must be utilized and no cement retained restoration margin should ever be more than 1 mm below the tissue.
9. All crowns and bridges must be fit to a solid cast to provide the most accurate proximal contacts.
10. Contract lab must be able to accept digital impressions.